Let's Practice

IF Statement

Example 1

```
always @ (*)
 begin
  if( done_flag == 2'b00)
    begin
      A = 1’b1 ;
      B = 1’b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1’b0 ;
      B = 1’b1 ;
    end
  else
    begin
      A = 1’b1 ;
      B = 1’b0 ;
    end
 end
```

No Unintentional Latch will be infered

Example 2

```
always @ (*)
  begin
  if(E)
    begin
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if (F)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b1 ;
    end
  else if (G)
    begin
      A = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Solution 1

```
always @ (*)
  begin
  if(E)
    begin
      A = 1'b0 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if (F)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b1 ;
    end
```

```
else if (G)
    begin
      A = 1'b0 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
end
```

Example 2

```
always @ (*)
  begin
  if(E)
    begin
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if (F)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b1 ;
    end
  else if (G)
    begin
      A = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Solution 2

```
always @ (*)
  begin
    A = 1'b0 ;
    B = 1'b0 ;
    C = 1'b0 ;
  if(E)
    begin
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if (F)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b1 ;
    end
```

```
else if (G)
    begin
      A = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Example 3

```
always @ (*)
  begin
  if( done_flag == 2'b00)
    begin
      A = 1’b1 ;
      B = 1’b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1’b0 ;
      B = 1’b1 ;
    end
  else if ( done_flag == 2'b11)
    begin
      A = 1’b1 ;
      B = 1’b0 ;
    end
end
```

Solution

```
always @ (*)
  begin
  if( done_flag == 2'b00)
    begin
      A = 1’b1 ;
      B = 1’b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1’b0 ;
      B = 1’b1 ;
    end
  else if ( done_flag == 2'b11)
    begin
      A = 1’b1 ;
      B = 1’b0 ;
    end
end
```

```
else
    begin
      A = 1’b0 ;
      B = 1’b0 ;
    end
end
```

Example 4

```
always @ (*)
  begin
  if( done_flag == 2'b00)
    begin
      A = 1’b1 ;
      B = 1’b0 ;
      C = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1’b0 ;
      B = 1’b1 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1’b1 ;
      B = 1’b0 ;
    end
end
```

Solution 1

```
always @ (*)
  begin
  if( done_flag == 2'b00)
    begin
      A = 1’b1 ;
      B = 1’b0 ;
      C = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1’b0 ;
      B = 1’b1 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1’b1 ;
      B = 1’b0 ;
      C = 1'b0 ;
    end
end
```

Example 4

```
always @ (*)
  begin
  if( done_flag == 2'b00)
    begin
      A = 1’b1 ;
      B = 1’b0 ;
      C = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1’b0 ;
      B = 1’b1 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1’b1 ;
      B = 1’b0 ;
    end
end
```

Solution 2

```
always @ (*)
 begin
 C = 1'b0 ;
  if( done_flag == 2'b00)
    begin
      A = 1’b1 ;
      B = 1’b0 ;
      C = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1’b0 ;
      B = 1’b1 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1’b1 ;
      B = 1’b0 ;
    end
end
```

Let's Practice

Case Statement

Example 1

```
always @ (*)
  case (sel)
   2'b00: begin
          A = 1'b1 ;
          B = 1'b0 ;
          C = 1'b1 ;
        end
   2'b01: begin
          A = 1'b1 ;
          B = 1'b0 ;
          C = 1'b1 ;
        end
   2'b10: begin
          A = 1'b1 ;
          B = 1'b0 ;
          C = 1'b1
        end
endcase
```

Solution

```
always @ (*)
  case (sel)
   2'b00: begin
          A = 1'b1 ;
          B = 1'b0 ;
          C = 1'b1 ;
        end
   2'b01: begin
          A = 1'b1 ;
          B = 1'b0 ;
          C = 1'b1 ;
        end
   2'b10: begin
          A = 1'b1 ;
          B = 1'b0 ;
          C = 1'b1
        end
  default: begin
          A = 1'b1 ;
          B = 1'b0 ;
          C = 1'b1
        end
endcase
```

Example 2

```
always @ (*)
  case (sel)
   2'b00: begin
            A = 1'b1 ;
            B = 1'b0 ;
            C = 1'b1 ;
          end
   2'b01: begin
            B = 1'b1 ;
          end
   2'b10: begin
            A = 1'b1 ;
          end
   2'b11: begin
            A = 1'b1 ;
            B = 1'b0 ;
          end
  endcase
```

Solution 1

```
always @ (*)
  case (sel)
   2'b00: begin
            A = 1'b1 ;
            B = 1'b0 ;
            C = 1'b1 ;
          end
   2'b01: begin
            A = 1'b1 ;
            C = 1'b0 ;
            B = 1'b1 ;
          end
   2'b10: begin
            A = 1'b1 ;
            B = 1'b0 ;
            C = 1'b1 ;
          end
   2'b11: begin
            A = 1'b1 ;
            B = 1'b0 ;
            C = 1'b1 ;
          end
  endcase
```

Example 2

```
always @ (*)
  case (sel)
   2'b00: begin
           A = 1'b1 ;
           B = 1'b0 ;
           C = 1'b1 ;
         end
   2'b01: begin
           B = 1'b1 ;
         end
   2'b10: begin
           A = 1'b1 ;
         end
   2'b11: begin
           A = 1'b1 ;
           B = 1'b0 ;
         end
  endcase
```

Solution 2

```
always @ (*)
 begin
   A = 1'b1 ;
   B = 1'b0 ;
   C = 1'b1 ;
    case (sel)
     2'b00: begin
             A = 1'b1 ;
             B = 1'b0 ;
             C = 1'b1 ;
            end
     2 'b01: begin
              B = 1'b1 ;
             end
     2'b10 : begin
              A = 1'b1 ;
             end
     2'b11 : begin
              A = 1'b1 ;
              B = 1'b0 ;
             end
     endcase
```

Example 3

```
always @ (*)
  begin
      A = 1'b1 ;
      B = 1'b0 ;
  case (sel)
   2'b00:  begin
             A = 1'b1 ;
             B = 1'b0 ;
             C = 1'b1 ;
            end
   2'b01:  begin
             B = 1'b1 ;
            end
   2'b10:  begin
             A = 1'b1 ;
            end
   default: begin
             A = 1'b1 ;
             B = 1'b0 ;
           end
  endcase
 end
```

Solution 1

```
always @ (*)
  begin
      A = 1'b1 ;
      B = 1'b0 ;
      C = 1'b1 ;
  case (sel)
   2'b00:  begin
             A = 1'b1 ;
             B = 1'b0 ;
             C = 1'b1 ;
            end
   2'b01:  begin
             B = 1'b1 ;
            end
   2'b10:  begin
             A = 1'b1 ;
            end
   default: begin
             A = 1'b1 ;
             B = 1'b0 ;
           end
  endcase
 end
```

Example 3

```
always @ (*)
  begin
      A = 1'b1 ;
      B = 1'b0 ;
  case (sel)
   2'b00:  begin
             A = 1'b1 ;
             B = 1'b0 ;
             C = 1'b1 ;
            end
   2'b01:  begin
             B = 1'b1 ;
            end
   2'b10:  begin
             A = 1'b1 ;
            end
   default: begin
             A = 1'b1 ;
             B = 1'b0 ;
           end
  endcase
 end
```

Solution 2

```
always @ (*)
  begin
      A = 1'b1 ;
      B = 1'b0 ;
   case (sel)
   2'b00:  begin
             A = 1'b1 ;
             B = 1'b0 ;
             C = 1'b1 ;
            end
   2'b01:  begin
             B = 1'b1 ;
             C = 1'b1 ;
            end
   2'b10:  begin
             A = 1'b1 ;
             C = 1'b1 ;
            end
   default: begin
                A = 1'b1 ;
                C = 1'b1 ;
               end
  endcase
```

Example 4

```
always @ (*)
 begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
 case (sel)
  2'b00: begin
          A = 1'b1 ;
        end
  2'b01: begin
          B = 1'b1 ;
        end
  2'b10: begin
          C = 1'b1 ;
        end
  default: begin
          A = 1'b1 ;
          B = 1'b0 ;
        end
 endcase
end
```

No Unintentional Latch will be infered